

The Natural Map to the Hyperseries Sequence Space Is Not Injective

A supermoderate-spike counterexample to the open problem in arXiv:2006.16141

Candidate full solution packet for arXiv:2006.16141

27 August 2026

Status. Candidate full negative answer, likely valid and awaiting human review. For every pair of gauges for which the source map is defined, its kernel is nonzero; in fact, the kernel contains infinitely many nonzero pairwise orthogonal idempotents.

1 Source question

The source is Diksha Tiwari and Paolo Giordano, *Hyperseries in the non-Archimedean ring of Colombeau generalized numbers*, arXiv:2006.16141, published in *Monatshefte für Mathematik* 197 (2022), 193–223 [1]. On source PDF page 7, immediately after Lemma 7(iii), the authors construct an ${}^\rho\widetilde{\mathbb{R}}$ -linear map

$$\lambda : {}^\rho\widetilde{\mathbb{R}}_{\mathbf{u}} \longrightarrow {}^\rho\widetilde{\mathbb{R}}_{\mathbf{s}}, \quad [a_{n\varepsilon}]_{\mathbf{u}} \longmapsto [a_{n\varepsilon}]_{\mathbf{s}},$$

under the gauge condition $\sigma \geq \rho^*$, and state:

“It remains an open problem the study of injectivity of the map λ .”

$$\left| \sum_{n=0}^{\lfloor \frac{1}{\varepsilon} \rfloor} a_{n\varepsilon} \right| \leq (\text{ni}(N)_{\varepsilon} + 1) \cdot \rho_{\varepsilon}^q$$

for ε small. Since $\text{ni}(N)_{\varepsilon}$ is σ -moderate and $\sigma \geq \rho^*$, this proves that the linear map λ is well-defined.

(iv): This follows directly from Def. 5. □

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We could say that $\sigma \geq \rho^*$ and $[a_{n\varepsilon}]_{\mathbf{u}} \in {}^\rho\widetilde{\mathbb{R}}_{\mathbf{u}}$ are sufficient conditions to get $(a_n)_n = [a_{n\varepsilon}]_{\mathbf{s}} \in {}^\rho\widetilde{\mathbb{R}}_{\mathbf{s}}$ and hence to start talking about hyperfinite sums $\sum_{n=N}^M a_n$ and hyperseries ${}^\rho\sum_{n \in {}^\sigma\widetilde{\mathbb{N}}} a_n$. On the other hand, if $\sigma \geq \rho^*$ is false, we can still consider the space ${}^\rho\widetilde{\mathbb{R}}_{\mathbf{s}}$ and hence talk about hyperseries, but the corresponding space ${}^\rho\widetilde{\mathbb{R}}_{\mathbf{s}}$ lacks elements such as $(1)_n$ because of the subsequent Lem. 10. As we will see in the following examples 8.3) and 8.6), the space ${}^\rho\widetilde{\mathbb{R}}_{\mathbf{s}}$ still contains sequences corresponding to interesting converging hyperseries.

The following examples of convergent hyperseries justify our definition of hyperseries by recovering classical examples such such geometric and exponential hyper-

Figure 1: The injectivity question in the source paper, source PDF page 7.

2 Definitions used

Let $\rho = (\rho_\varepsilon)$ and $\sigma = (\sigma_\varepsilon)$ be gauges, so in particular $\rho_\varepsilon, \sigma_\varepsilon > 0$ and both tend to zero. The condition $\sigma \geq \rho^*$ means that

$$\sigma_\varepsilon \geq \rho_\varepsilon^Q \quad \text{eventually for some } Q \in \mathbb{N}_{>0}.$$

An array $(a_{n\varepsilon})_{n \in \mathbb{N}, \varepsilon}$ is *uniformly moderate* if

$$\exists Q \in \mathbb{N} \forall \varepsilon \forall n \in \mathbb{N} : |a_{n\varepsilon}| \leq \rho_\varepsilon^{-Q},$$

and is *uniformly negligible* if

$$\forall q \in \mathbb{N} \forall \varepsilon \forall n \in \mathbb{N} : |a_{n\varepsilon}| \leq \rho_\varepsilon^q.$$

The quotient is ${}^\rho\widetilde{\mathbb{R}}_u$.

Every $N = [N_\varepsilon] \in {}^\sigma\widetilde{\mathbb{N}}$ has a σ -moderate representative: for some $R \in \mathbb{N}$,

$$N_\varepsilon \leq \sigma_\varepsilon^{-R} \quad \text{eventually.}$$

The target ${}^\rho\widetilde{\mathbb{R}}_s$ identifies two arrays if every interval sum between σ -moderate hypernatural endpoints is ρ -negligible. Thus an array represents zero in ${}^\rho\widetilde{\mathbb{R}}_s$ precisely when, for all $M, N \in {}^\sigma\widetilde{\mathbb{N}}$,

$$\left(\sum_{n=N_\varepsilon}^{M_\varepsilon} a_{n\varepsilon} \right)_\varepsilon \sim_\rho 0.$$

These are the definitions in Section 2 of [1] specialized to the zero class.

3 Proof intuition

The two quotient relations observe different ranges of the ordinary index n . Uniform negligibility in ${}^\rho\widetilde{\mathbb{R}}_u$ tests *every* $n \in \mathbb{N}$ at each small ε . In contrast, ${}^\rho\widetilde{\mathbb{R}}_s$ only tests intervals whose endpoints grow at most like a fixed power of $1/\sigma_\varepsilon$.

Place a single unit spike at an index K_ε growing faster than every such power. The domain still sees the spike, because its definition takes a supremum over all ordinary n . Every admissible hyperseries interval misses it eventually, so the target sees the zero array. The counterexample is the gap between “all ordinary indices” and “all σ -moderate indices” made literal.

4 Formal result

Lemma 1 (A super- σ index). *Define*

$$K_\varepsilon := \lceil \exp(1/\sigma_\varepsilon) \rceil.$$

Then $K_\varepsilon > \sigma_\varepsilon^{-R}$ eventually for every fixed $R \in \mathbb{N}$. Consequently, if $N = [N_\varepsilon] \in {}^\sigma\widetilde{\mathbb{N}}$, then $N_\varepsilon < K_\varepsilon$ eventually.

Proof. Put $x_\varepsilon = 1/\sigma_\varepsilon$. Then $x_\varepsilon \rightarrow +\infty$, and for every fixed R ,

$$\frac{\exp(1/\sigma_\varepsilon)}{\sigma_\varepsilon^{-R}} = \frac{e^{x_\varepsilon}}{x_\varepsilon^R} \longrightarrow +\infty.$$

This proves the first assertion. For the second, choose R such that $N_\varepsilon \leq \sigma_\varepsilon^{-R}$ eventually and apply the first assertion. $\square$

Theorem 1 (Complete negative answer to the injectivity problem). *For every pair of gauges ρ, σ satisfying $\sigma \geq \rho^*$, the natural linear map*

$$\lambda : {}^\rho\widetilde{\mathbb{R}}_{\mathbf{u}} \longrightarrow {}^\rho\widetilde{\mathbb{R}}_{\mathbf{s}}$$

of Lemma 7(iii) in [1] is not injective.

Proof. Let K_ε be as in Lemma 1 and define

$$a_{n\varepsilon} := \begin{cases} 1, & n = K_\varepsilon, \\ 0, & n \neq K_\varepsilon. \end{cases}$$

Because $|a_{n\varepsilon}| \leq 1 = \rho_\varepsilon^0$, this array is uniformly moderate. It is not uniformly negligible: since $\rho_\varepsilon \rightarrow 0$, eventually $\rho_\varepsilon < 1$, while for the ordinary index $n = K_\varepsilon$ one has $|a_{K_\varepsilon, \varepsilon}| = 1 > \rho_\varepsilon$. Therefore

$$[a_{n\varepsilon}]_{\mathbf{u}} \neq 0 \quad \text{in } {}^\rho\widetilde{\mathbb{R}}_{\mathbf{u}}.$$

Now fix arbitrary $M = [M_\varepsilon]$ and $N = [N_\varepsilon]$ in ${}^\sigma\widetilde{\mathbb{N}}$. By Lemma 1, both endpoints are eventually strictly smaller than K_ε . Hence every ordinary integer between the two endpoints is also smaller than K_ε , and

$$\sum_{n=N_\varepsilon}^{M_\varepsilon} a_{n\varepsilon} = 0 \quad \text{eventually.}$$

The interval-sum net is therefore ρ -negligible. Since M, N were arbitrary, $(a_{n\varepsilon})$ represents zero in ${}^\rho\widetilde{\mathbb{R}}_{\mathbf{s}}$. Thus

$$\lambda([a_{n\varepsilon}]_{\mathbf{u}}) = [a_{n\varepsilon}]_{\mathbf{s}} = 0 \quad \text{while} \quad [a_{n\varepsilon}]_{\mathbf{u}} \neq 0,$$

which proves noninjectivity. $\square$

Corollary 1 (A large invisible tail). *The kernel of λ contains infinitely many nonzero pairwise orthogonal idempotents of the pointwise ring ${}^\rho\widetilde{\mathbb{R}}_{\mathbf{u}}$.*

Proof. For $j \in \mathbb{N}_{>0}$ set

$$K_\varepsilon^{(j)} = \lceil e^{j/\sigma_\varepsilon} \rceil, \quad a_{n\varepsilon}^{(j)} = \mathbf{1}_{\{K_\varepsilon^{(j)}\}}(n).$$

The proof of Theorem 1 applies to each $a^{(j)}$. Each domain class is nonzero and idempotent. If $j \neq k$, then $K_\varepsilon^{(j)} \neq K_\varepsilon^{(k)}$ eventually, so the pointwise product of the two arrays is eventually identically zero, uniformly in n . Their classes are therefore orthogonal. $\square$

5 Verification notes

- **Gauge generality.** No special form, monotonicity, or comparison beyond the source hypothesis $\sigma \geq \rho^*$ is used. In fact, that comparison is needed only so that the source map λ is defined.
- **Domain nonzero.** The witness to failure of uniform negligibility is allowed to depend on ε : the quantifier order is $\forall^0\varepsilon \forall n$. At each ε , choosing the ordinary integer $n = K_\varepsilon$ contradicts the $q = 1$ estimate.
- **Target zero.** Every individual hypernatural has some fixed moderateness exponent. After taking the maximum exponent for two endpoints, Lemma 1 puts the spike beyond the entire interval.
- **Representatives.** The proof checks the target equivalence for arbitrary hypernatural classes by choosing arbitrary moderate representatives. Changing representatives does not affect the conclusion, since the source definition already makes interval sums well defined.
- **No computation.** The argument is exact and uses only $e^x/x^R \rightarrow \infty$; no numerical or symbolic dependency remains.

6 Scope and novelty check

The theorem answers the printed injectivity problem completely and negatively for every gauge pair in the stated domain of the map. It does not alter the usefulness of λ for transferring convergence statements; it identifies an “invisible tail” that all σ -moderate hyperseries intervals fail to detect.

A bounded novelty check was performed on 27 August 2026. The four local run indexes were searched for arXiv:2006.16141 and the phrases “hyperseries”, “injectivity of the map λ ”, “uniformly moderate”, and “supermoderate”. Exact-phrasal web searches found the published source, which still states the problem open, and a 2025 University of Vienna thesis that reproduces the same open statement. The local arXiv sources of the follow-up *Hyper-power series and generalized real analytic functions* (arXiv:2212.04757) [2], as well as citing arXiv sources 2111.15408, 2501.17364, and 2604.17904, were searched for “injectiv”, “ λ ”, and the two quotient-space notations; no resolution was found. No paper or packet containing this supermoderate-spike counterexample was located. The search is bounded rather than exhaustive, so novelty confidence is moderate to high.

7 Human-review recommendation

Promote as a full negative answer if a reviewer confirms the quantifier order in uniform negligibility and the standard definition that every element of ${}^\sigma\tilde{\mathbb{N}}$ has a representative bounded by one fixed power of $1/\sigma_\varepsilon$. Those are stated explicitly in the source. The remaining argument is elementary, works for arbitrary admissible gauges, and has no unproved dependency.

References

- [1] D. Tiwari and P. Giordano, *Hyperseries in the non-Archimedean ring of Colombeau generalized numbers*, Monatshefte für Mathematik **197** (2022), 193–223; arXiv:2006.16141, doi:10.1007/s00605-021-01647-0.
- [2] D. Tiwari, A. Mukhammadiev, and P. Giordano, *Hyper-power series and generalized real analytic functions*, arXiv:2212.04757 (2022).